# A PROPOSITION FOR FIRST PROOF: SOLUTION

Problem statement.

Suppose $(M, g)$ and $(N, h)$ are piecewise smooth Riemannian manifolds. If $f : M \to N$ is piecewise smooth, we define the $k$-dilation of $f$ as the infimal $\Lambda$ so that for every $k$-dimensional submanifold $\Sigma \subset M$,

$$\mathrm{Vol}_k(f(\Sigma)) \leq \Lambda \mathrm{Vol}_k(\Sigma).$$

We write the $k$-dilation of $f$ as $\mathrm{Dil}_k(f)$. Equivalently, $\mathrm{Dil}_k(f) = \sup_{x \in M} \|\Lambda^k df\|$.

**Proposition 0.1.** *Prove that there is a constant $k > 0$ so that the following holds.*

*Suppose that $R \subset \mathbb{R}^4$ is a 4-dimensional rectangle with side lengths $R_1 \leq R_2 \leq R_3 \leq R_4$ and $S$ is a 4-dimensional rectangle with side lengths $S_1 \leq S_2 \leq S_3 \leq S_4$. Suppose that $f : (R, \partial R) \to (S, \partial S)$ is a piecewise smooth map with degree 1 and with $Dil_2(f) \leq 1$. Suppose that $R_1 \leq kS_1$. Then either*

$$R_3 R_4 > k S_3 S_4, or$$

$$R_1 R_2 R_3 R_4 > k S_1 S_2^{1/2} S_3^{3/2} S_4.$$

We write $A \lesssim B$ to mean that there is some positive constant $C$ so that $A \leq CB$. We write $A \sim B$ to mean that $A \lesssim B$ and $B \lesssim A$.

0.1. **Isoperimetric lemma.** The first main ingredient of our proof is an isoperimetric inequality for relative cycles in a rectangle. Thie lemma appeared in [3] as Theorem 3, but since [3] was not published we include the full proof here.

Let $R$ denote the n-dimensional rectangle $[0, R_1] \times ... \times [0, R_n]$, where the dimensions are ordered so that $R_1 \leq ... \leq R_n$. If $z$ is a relative integral k-cycle in $R$, the filling volume of $z$ is the smallest volume of any relative (k+1)-chain $y$ with $\partial y = z$. Let $I_R^k(V)$ denote the largest filling volume of any k-dimensional relative integral cycle in $R$ with volume at most $V$.

Remark: We use the following definition for volume. If a chain $C$ is given by $\sum c_i f_i$ where $c_i \in \mathbb{Z}$ and $f_i$ is a Lipschitz map from the standard k-simplex to $S$, then the volume of the chain $C$ is defined to be $\sum |c_i| Vol(f_i^* Euc)$, where $Euc$ denotes the Euclidean metric on $R$. This quantity is also called the mass of $C$.

**Lemma 0.1.** $I_R^0(V) \leq R_1 V$

*Proof.* When $k = 0$, $z$ is just a weighted sum of points $\sum c_i p(i)$, where $c_i \in \mathbb{Z}$ and $p(i)$ is a point in $R$. The volume of $z$ is defined to be $\sum |c_i|$. A point $p$ with coordinates $(p_1, ..., p_n)$ bounds a segment $[0, p_1] \times \{p_2\} \times ... \times \{p_n\}$. Applying this operation to each point $p(i)$ with multiplicity $c_i$, we get a filling of $z$ with volume at most $R_1 Vol_0(z)$. $\square$

**Lemma 0.2.** *(Isoperimetric lemma) There are constants $c(n) > 0, C(n)$ so that the following holds for any $1 \leq k \leq n - 1$.*

*If $V \leq c(n) R_1 ... R_k$, then write $V = c(n) R_1 ... R_j \rho^{k-j}$ for some $0 \leq j \leq k-1$ and some $\rho$ in the range $R_j \leq \rho \leq R_{j+1}$. (These conditions determine $j$ and $\rho$ uniquely.)*

$$\textit{Then } I_R^k(V) \leq C(n) R_1 ... R_j \rho^{k-j+1}.$$

*In any case, $I_R^k(V) \leq C(n) R_{k+1} V$.*

The proof of the isoperimetric lemma is based on using the Federer-Fleming deformation theorem at multiple scales. We first recall a version of the Federer-Fleming deformation theorem.

**Theorem 1.** *(Federer-Fleming) Suppose that $z$ is a relative k-cycle in $R$. Consider a rectangular lattice inside $R$ with each side-length roughly equal to $L$ (up to a factor of 2), for some $L \leq R_1$, and suppose that the boundary of $R$ lies in the (n-1)-skeleton of the lattice. Then there is another relative cycle $z'$ in $R$ contained in the k-skeleton of the lattice and obeying the following inequalities.*

*1. The volume of $z'$ is at most $C(n)|z|$.*

*2. The filling volume of $z' - z$ is at most $C(n) L |z|$.*

See [2] for a proof of this form of the Federer-Fleming deformation theorem.

*Proof of Lemma 0.2.* We prove the isoperimetric inequality by induction on $k$. Lemma 0.1 serves as a base case.

If $k \geq 2$, suppose by induction that the Lemma holds for $k-1$.

Suppose that $z$ is a k-cycle with volume $V$. We proceed in two cases. If $V \leq c(n) R_1^k$, then we select $L = C(n) V^{1/k} \leq R_1$ and pick a rectangular lattice with sidelengths roughly $L$ and with $\partial R$ in the (n-1)-skeleton of the lattice. Then we use Lemma 1.1 to move $z$ to a new relative cycle $z'$ with volume at most $C(n)V$ lying in the k-skeleton of our lattice. Since $C(n) V \leq L^k$, the new cycle $z'$ is simply 0. Lemma 1.1 also guarantees us a homology from $z$ to $z'$ with volume at most $C(n) L V$, which is at most $C(n) V^{\frac{k+1}{k}}$. This upper bound is the one we needed to prove.

In the second case, we suppose that $V \geq c(n) R_1^k$. In this case, we select $L = R_1$ and pick a rectangular lattice with sidelengths roughly $L$ and with $\partial R$ in the (n-1)-skeleton of the lattice. We pick the lattice so that each lattice point has $x_1$ coordinate either 0 or $R_1$. Then we use Lemma 1.1 to move $z$ to a new relative cycle $z'$ with volume at most $C(n)V$ lying in the k-skeleton of our lattice. The homology from $z$ to $z'$ has volume at most $C(n) R_1 V$. The cycle $z'$ need not be 0, but it is a union of interior k-faces of our lattice. Each interior k-face has the form $[0, R_1] \times ...$, and so the cycle $z'$ has the special form $z' = [0, R_1] \times z_1$ for some relative cycle $z_1$ in the (n-1)-dimensional rectangle $[0, R_2] \times ... \times [0, R_n]$. The cycle $z_1$ has volume at most $C(n) V / R_1$.

If $k = 1$, then we find a filling of $z_1$ using Lemma 0.1, and if $k > 1$, then we find a filling of $z_1$ by induction. Therefore, $z_1$ bounds a relative chain $C_1$ with a certain volume bound that we calculate below. Then $z'$ bounds $[0, R_1] \times C_1$. We will calculate that the volume of this filling obeys the inequality stated in the theorem.

If $k = 1$, then we get the bound

$$\mathrm{Vol}_2([0, R_1] \times C_1) = R_1 \mathrm{Vol}_1(C_1) \leq R_1 R_2 \mathrm{Vol}_0(z_1) \leq C(n) R_2 V,$$

which is the desired bound in the case that $V \geq c(n) R_1$.

Now we turn to the case $k > 1$.

If the volume of $z$ is at most $c(n)R_1...R_k$, then the volume of $z_1$ is at most $c(n-1)R_2...R_k$. If the volume of $z$ is equal to $c(n)R_1...R_j\rho^{k-j}$ for some $\rho$ in the range $R_j \le \rho \le R_{j+1}$, then the volume of $z_1$ is roughly $R_2...R_j\rho^{(k-1)-(j-1)}$ for the same $\rho$. By induction, $z_1$ bounds a chain $C_1$ with volume at most $C(n-1)R_2...R_j\rho^{k-j+1}$, and so $[0, R_1] \times C_1$ has volume at most $C(n-1)R_1...R_j\rho^{k-j+1}$. Also, the homology from $z$ to $z'$ has volume at most $C(n)R_1(R_1...R_j)\rho^{(k-j)} \le C(n)R_1...R_j\rho^{k-j+1}$. Therefore, the filling volume of $z$ is at most $C(n)R_1...R_j\rho^{k-j+1}$. This gives the desired bound if the volume of $z$ is at most $c(n)R_1...R_k$.

Regardless of the volume of $z$, $z_1$ bounds a k-chain $C_1$ of volume at most $C(n-1)R_{k+1}V/R_1$. Hence $z' = [0, R_1] \times z_1$ bounds a (k+1)-chain of volume at most $C(n-1)R_{k+1}V$. The homology from $z$ to $z'$ has volume at most $C(n)R_1V$. Therefore, the filling volume of $z$ is at most $C(n)R_{k+1}V$. □

0.2. **Linking invariants.** An important character in the proof will be a certain homotopy invariant that we call a linking invariant. We now describe this invariant and review standard properties.

Consider the pair $(S^2 \vee S^2, *)$, where $*$ is the wedge point of $S^2 \vee S^2$. Let $\Omega_1, \Omega_2$ be generators of $H^2(S^2 \vee S^2, *)$ corresponding to the two copies of $S^2$, and let $\omega_1, \omega_2$ be differential forms representing $\Omega_1, \Omega_2$. We choose $\omega_i$ so that it is supported in the $i^{th}$ $S^2$ outside of a neighborhood of $*$.

Suppose that $(M, \partial M)$ is a compact manifold with boundary, and $h \in H_3(M, \partial M; \mathbb{Z})$. Suppose that $g : (M, \partial M) \to (S^2 \vee S^2, *)$ is a piecewise smooth map with $g^*(\Omega_i) = 0$ in $H^2(M, \partial M; \mathbb{Z})$ for $i = 1, 2$. By the de Rham theorem for cohomology with compact support, it follows that there are piecewise smooth differential forms $\alpha_1, \alpha_2$ on $(M, \partial M)$ with

- $\alpha_i$ is supported in interior of $M$.
- $d\alpha_i = g^*(\omega_i)$

If $z$ is a relative 3-chain in $(M, \partial M)$ with $[z] = h$, we define

$$L_h(g) := \int_z \alpha_1 \wedge g^*(\omega_2) \tag{0.1}$$

If $(M, \partial M)$ is an oriented 3-manifold with fundamental homology class $[M]$, we abbreviate $L(g) = L_{[M]}(g)$.

**Lemma 0.3.** *Assuming that $g^*(\Omega_i) = 0 \in H^2(M, \partial M)$ for $i = 1, 2$, $L_h$ obeys the following properties.*

1. *$L_h(g)$ is independent of the choice of $z$*
2. *If $\beta_1$ is a 1-form on $(M, \partial M)$ with $d\beta_1 = g^*(\omega_1)$, even if the support of $\beta_1$ includes the boundary of $M$, we still have*
$$L_h(g) = \int_z \beta_1 \wedge g^*(\omega_2).$$
3. *$L_h(g)$ is a homotopy invariant of $g$.*
4. *If $\phi : (L, \partial L) \to (M, \partial M)$ and $\phi_*(h_L) = h_M$, then*
$$L_{h_L}(g \circ \phi) = L_{h_M}(g).$$

*Proof.* (1) Note that $d(\alpha_1 \wedge g^*(\omega_2)) = g^*(\omega_1) \wedge g^*(\omega_2) = g^*(\omega_1 \wedge \omega_2) = 0$. If $[z_1] = [z_2] = h$, then $z_1 - z_2 = \partial y$ for a relative 4-chain $y$, and the equality follows from Stokes theorem.

(2) If $d\beta_1 = d\alpha_1$, then $\int_z \beta_1 \wedge g^*(\omega_2) - \int_z \alpha_1 \wedge g^*(\omega_2) = \int_z (\alpha_1 - \beta_1) \wedge d\alpha_2 = \int_z d\left((\alpha_1 - \beta_1) \wedge \alpha_2\right)$. Apply Stokes theorem and note that $(\alpha_1 - \beta_1) \wedge \alpha_2$ vanishes on $\partial z \subset \partial M$ because $\alpha_2$ vanishes there.

(3) Suppose $G : (M \times [0,1], \partial M \times [0,1]) \to (S^2 \vee S^2, *)$ is a homotopy between $g_0$ and $g_1$. Let $Z = z \times [0,1]$. Choose $A_i$ to be forms on $(M \times [0,1], \partial M \times [0,1])$, supported in $Int(M) \times [0,1]$ with $dA_i = G^*(\omega_i)$. We have $d(A_1 \wedge G^*(\omega_2)) = 0$ as in item 1, and so by Stokes theorem on $Z$,

$$L_h(g_0) = \int_{z \times \{0\}} A_1 \wedge G^*(\omega_2) = \int_{z \times \{1\}} A_1 \wedge G^*(\omega_2) = L_h(g_1).$$

(4) Choose $\alpha_{i,M}$ supported on $Int(M)$ so $d\alpha_{i,M} = g^*(\omega_i)$. Then define $\alpha_{i,L} = \phi^*(\alpha_{i,M})$, supported in $Int(L)$ with $d\alpha_{i,L} = (g \circ \phi)^*(\omega_2)$. Choose $z_L$ with $[z_L] = h_L$, and set $z_M = \phi(z_L)$ so $[z_M] = h_M$. Then

$$L_{h_L}(g \circ \phi) = \int_{z_L} \alpha_{1,L} \wedge (g \circ \phi)^*(\omega_2) = \int_{z_M} \alpha_{1,M} \wedge g^*(\omega_2) = L_{h_M}(g).$$

□

This linking invariant is not always zero. Let $Q_1$ be the unit 3-cube. There is a piecewise smooth map $g_1 : (Q_1, \partial Q_1) \to (S^2 \vee S^2, *)$ with $L(g_1) = 1$. This is a standard construction. For details, see Proposition 5.2 in [4].

We equip $(S^2 \vee S^2, *)$ with a metric where each $S^2$ is a round sphere of radius $S_1$. Using this metric, we can choose the differential forms $\omega_i$ so that

$$\|\omega_i\|_{L^\infty} \lesssim S_1^{-2}. \tag{0.2}$$

Let $Q_{S_1}$ be a 3-cube of side length $S_1$. By scaling $g_1$, we can construct a map $g_{S_1} : (Q_{S_1}, \partial Q_{S_1}) \to (S^2 \vee S^2, *)$ with $\mathrm{Dil}_1(g_{S_1}) \lesssim 1$ and $L(g_{S_1}) = 1$.

The connection between $k$-dilation and linking invariants such as this one is explored in [4].

### 0.3. **The proof of the Proposition.**

*Proof.* For each $(a,b) \in [0, S_3] \times [0, S_4]$, let $\Sigma_{a,b} := [0, S_1] \times [0, S_2] \times \{a\} \times \{b\} \subset S$. By the transversality theorem, for almost every $(a,b) \in [0, S_3] \times [0, S_4]$, the map $f$ is transverse to $\Sigma_{a,b}$ and so $f^{-1}(\Sigma_{a,b})$ is a piecewise smooth submanifold.

Define a set $A$ of good values of $(a,b)$ by

$$A := \{(a,b) \in [(1/4)S_3, (3/4)S_3] \times [(1/4)S_4, (3/4)S_4] : f \text{ is transverse to } \Sigma_{a,b}\}. \tag{0.3}$$

We will prove a lower bound for $\mathrm{Vol}_2(\Sigma_{a,b})$ for each $(a,b) \in A$. Note that the measure of $A$ is $|A| \sim S_3 S_4$.

Next we discuss the topology of the pair $(S, \partial S \cup \Sigma_{a,b})$. Note that $H_2(\partial S \cup \Sigma_{a,b}; \mathbb{Z}) = \mathbb{Z}$. To build a generator, note that $\partial \Sigma_{a,b}$ is homologically trivial in $\partial S$, and let $y$ be a 2-chain in $\partial S$ with $\partial y = -\partial \Sigma_{a,b}$. Then let $z_{2,S} = y + \Sigma_{a,b}$, which is a cycle in $\partial S \cup \Sigma_{a,b}$ representing a generator of $H_2(\partial S \cup \Sigma_{a,b})$, which we call $h_{2,S}$.

Also note that $H_3(S, \partial S \cup \Sigma_{a,b}; \mathbb{Z}) = \mathbb{Z}$. To build a generator, let $z_{3,S}$ be a relative 3-chain in $(S, \partial S)$ with $\partial z_{3,S} = \Sigma_{a,b}$. Then $z_{3,S}$ is a relative 3-cycle in $(S, \partial S \cup \Sigma_{a,b})$ representing a generator of $H_3(S, \partial S \cup \Sigma_{a,b}; \mathbb{Z})$, which we call $h_{3,S}$. Note that $\partial h_{3,S} = h_{2,S}$.

Next we build a topologically interesting map

$$g : (S, \partial S \cup \Sigma_{a,b}) \to (S^2 \vee S^2, *).$$

Define $S' = [0, S_1] \times [0, S_2] \times [0, S_3]$. Let $r : [0, S_3] \times [0, S_4] \to [0, S_3]$ be given by

$$r(y_3, y_4) = \min(10(|y_3 - a| + |y_4 - b|), S_3).$$

Now let $g_1(y_1, y_2, y_3, y_4) = (y_1, y_2, r(y_3, y_4))$ so $g_1 : S \to S'$. Notice that $g_1(\Sigma_{a,b}) \subset [0, S_1] \times [0, S_2] \times \{0\} \subset \partial S'$ and by (0.3), $g_1(\partial S) \subset \partial S'$. Therefore $g_1$ is a map of pairs $g_1 : (S, \partial S \cup \Sigma_{a,b}) \to (S', \partial S')$. Also $g_{1,*}(h_{3,S})$ is the fundamental homology class of $(S', \partial S')$.

Suppose that $Q_1, ..., Q_N \subset S'$ are disjoint cubes of side length $S_1$, with $N \sim \frac{S_2 S_3}{S_1^2}$. For each cube $Q$, recall that we defined above a map $g_Q : (Q, \partial Q) \to (S^2 \vee S^2, *)$ with $L(g_Q) = 1$ and $\mathrm{Dil}_1(g_Q) \lesssim 1$. Now we define $g_2$ so that for each $Q$, $g_2|_Q = g_Q$, and so that $g_2(x) = *$ if $x$ is not in any $Q$. Then $g_2 : (S', \partial S') \to (S^2 \vee S^2, *)$ has $\mathrm{Dil}_1(g_2) \lesssim 1$.

Since $H^2(S', \partial S') = 0$, $g_2^*(\Omega_i) = 0$ for $i = 1, 2$ and $L(g_2)$ is well defined. Next we compute $L(g_2)$. Since $g_2|_Q = g_Q$, we see that $g_2^*(\omega_i)|_Q = g_Q^*(\omega_i)$. For each $Q$, choose $\alpha_{1,Q}$ with compact support in $Int(Q)$ so that $d\alpha_{1,Q} = g_Q^*(\omega_1)$. Then define $\alpha_{1,2}$ so that $\alpha_{1,2}|_Q = \alpha_{1,Q}$ for each $Q$ and $\alpha_{1,2}$ vanishes outside of the union of the cubes $Q$. Then

$$L(g_2) = \int_{S'} \alpha_{1,2} \wedge g_2^*(\omega_2) = \sum_Q \int_Q \alpha_{1,Q} \wedge g_Q^*(\omega_2) = \sum_Q L(g_Q) = \sum_{Q \in S'} 1 = N \sim \frac{S_2 S_3}{S_1^2}.$$

Now we define $g = g_2 \circ g_1 : (S, \partial S \cup \Sigma_{a,b}) \to (S^2 \vee S^2, *)$. We have $\mathrm{Dil}_1(g) \lesssim 1$ and so $\mathrm{Dil}_2(g) \lesssim 1$. Since $g_{1,*}(h_{3,S}) = [S']$, we have

$$L_{h_{3,S}}(g) = L(g_2) \sim \frac{S_2 S_3}{S_1^2}. \tag{0.4}$$

For each $(a, b) \in A$, we view the map $f$ as a map of pairs $f : (R, \partial R \cup f^{-1}(\Sigma_{a,b})) \to (S, \partial S \cup \Sigma_{a,b})$. Recall that for $(a, b) \in A$, $f$ is transverse to $\Sigma_{a,b}$ and so $f^{-1}(\Sigma_{a,b})$ is a piecewise smooth submanifold.

We can define a homology class $h_{2,R} \in H_2(R, \partial R \cup f^{-1}(\Sigma_{a,b}); \mathbb{Z})$ just as we defined $h_{2,S}$: find a 2-chain $y$ in $\partial R$ with $\partial y = -\partial f^{-1}(\Sigma_{a,b})$ and set $z_{2,R} = y + f^{-1}(\Sigma_{a,b})$. Since $f : (R, \partial R) \to (S, \partial S)$ is degree 1, it follows that $f : (f^{-1}(\Sigma_{a,b}), \partial f^{-1}(\Sigma_{a,b})) \to (\Sigma_{a,b}, \partial \Sigma_{a,b})$ is also degree 1, and so $f_*(h_{2,R}) = h_{2,S}$.

Now suppose that $z_3$ is a relative 3-chain in $(R, \partial R)$ with $\partial z_3 = f^{-1}(\Sigma_{a,b})$. We can also view $z_3$ as a relative 3-cycle in $(R, \partial R \cup f^{-1}(\Sigma_{a,b}))$. We let $h_{3,R} = [z_3]$. Then we have $\partial h_{3,R} = h_{2,R}$.

If we view $f$ as a map of pairs, $f : (R, \partial R \cup f^{-1}(\Sigma_{a,b})) \to (S, \partial S \cup \Sigma_{a,b})$, then $f_*(h_{3,R}) = h_{3,S}$. To see this, note that $\partial f_*(h_{3,R}) = f_*(h_{2,R}) = h_{2,S}$. But the $\partial$ map $H_3(S, \partial S \cup \Sigma_{a,b}) \to H_2(\partial S \cup \Sigma_{a,b})$ is an isomorphism, and the only class $h$ with $\partial h = h_{2,S}$ is $h = h_{3,S}$.

Now we consider the map $F = g \circ f : (R, \partial R \cup f^{-1}(\Sigma_{a,b})) \to (S^2 \vee S^2, *)$. Since $\mathrm{Dil}_1(g) \lesssim 1$, we have $\mathrm{Dil}_2(g) \lesssim 1$ and so $\mathrm{Dil}_2(F) \lesssim 1$. By Lemma 0.3, we also have

$$L_{h_{3,R}}(F) = L_{h_{3,S}}(g) \sim \frac{S_2 S_3}{S_1^2}. \tag{0.5}$$

Next we give an upper bound for $L_{h_{3,R}}(F)$. Recall that $z_3$ is any relative 3-chain in $(R, \partial R)$ with $\partial z_3 = f^{-1}(\Sigma_{a,b})$, which we can also view as a relative 3-cycle in $(R, \partial R \cup f^{-1}(\Sigma_{a,b}))$ with $[z_3] = h_{3,R}$.

Recall that $\|\omega_i\|_{L^\infty} \lesssim S_1^{-2}$. Since $\mathrm{Dil}_2(F) \lesssim 1$, we have

$$\|F^*(\omega_i)\| \lesssim S_1^{-2} \tag{0.6}$$

Next we choose $\beta_1$ a 1-form on $R$ with $d\beta_1 = F^*(\omega_1)$. We choose $\beta_1$ by integrating $F^*(\omega_1)$ in the $x_1$ direction. Let $(x_1, x_2, x_3, x_4)$ be the standard coordinates on $R = [0, R_1] \times ... \times [0, R_4]$. If we write

$$F^*(\omega_1) = \sum_{j=2}^{4} c_j(x) dx_1 \wedge dx_j + \sum_{2 \le j_1 < j_2 \le 4} c_{j_1, j_2}(x) dx_{j_1} \wedge dx_{j_2},$$

then

$$\beta_1 = \sum_{j=2}^{4} \left( \int_0^{x_1} c_j(x_1', x_2, x_3, x_4) dx_1' \right) dx_j.$$

Since $F^*(\omega_1)$ is closed, it is a standard fact that $d\beta_1 = F^*(\omega_1)$ – see the proof of the Poincare lemma in Chapter 1 of [1]. We have $\|c_j\|_\infty \le \|F^*(\omega_i)\|_{L^\infty} \lesssim S_1^{-2}$, and $0 \le x_1 \le R_1$, and so

$$\|\beta_1\|_\infty \lesssim R_1 S_1^{-2} \tag{0.7}$$

Now we have

$$|L_{h_{3,R}}(F)| = \left| \int_{z_3} \beta_1 \wedge F^*(\omega_2) \right| \lesssim \mathrm{Vol}_3(z_3) R_1 S_1^{-4}.$$

Combining this with (0.5) and rearranging, we get

$$R_1^{-1} S_1^2 S_2 S_3 \lesssim \mathrm{Vol}_3(z_3). \tag{0.8}$$

Next we upper bound $\mathrm{Vol}_3(z_3)$ using the isoperimetric lemma, Lemma 0.2. For reference, we record here the relevant special cases of Lemma 0.2.

**Lemma 0.4.** *There are constants $c > 0, C$ so that the following holds.*

*If $y$ is a relative 2-cycle in $R$ with*

$$cR_1^2 \le Vol_2(y) \le cR_1 R_2, \tag{0.9}$$

*then $y = \partial z$ for a relative 3-chain $z$ in $R$ with*

$$Vol_3(z) \le CR_1^{-1} Vol_2(y)^2. \tag{0.10}$$

*If $y$ is a relative 2-cycle in $R$ with*

$$Vol_2(y) \leq cR_1^2, \tag{0.11}$$

*then $y = \partial z$ for a relative 3-chain $z$ in $R$ with*

$$Vol_3(z) \leq C\,Vol_2(y)^{3/2} \leq CR_1^3. \tag{0.12}$$

We apply this theorem with $y = f^{-1}(\Sigma_{a,b})$ and $z = z_3$. Since $R_1 \leq kS_1$, we have $\mathrm{Vol}_3(z_3) \gtrsim R_1^{-1}S_1^2S_2S_3 \geq k^{-3}R_1^3$. By choosing $k > 0$ small enough, we can guarantee that (0.12) does not hold. Examining Lemma 0.4, we see that only two possibilities remian:for every $(a,b) \in A$, either

$$\mathrm{Vol}_2(f^{-1}(\Sigma_{a,b})) \gtrsim R_1R_2, or \tag{0.13}$$

$$\mathrm{Vol}_2[f^{-1}(\Sigma_{a,b})] \gtrsim R_1^{1/2}\mathrm{Vol}_3(z_3)^{1/2} \gtrsim S_1S_2^{1/2}S_3^{1/2} \tag{0.14}$$

Now using $\mathrm{Dil}_2(f) \leq 1$ and the coarea inequality, we have

$$R_1R_2R_3R_4 = \mathrm{Vol}_4(R) \geq \int_A \mathrm{Vol}_2(f^{-1}(\Sigma_{a,b}))dadb.$$

Using $|A| \sim S_3S_4$ and plugging in the bound from (0.13) and (0.14), we see that

$$R_1R_2R_3R_4 \gtrsim R_1R_2S_3S_4, or$$

$$R_1R_2R_3R_4 \gtrsim S_1S_2^{1/2}S_3^{3/2}S_4.$$

□

## References

[1] R. Bott and L. Tu, *Differential Forms in Algebraic Topology*
[2] Guth, L., Notes on Gromov's systolic estimate, Geometriae Dedicata 123, Number 1, Dec. 2006, 113-129.
[3] Guth, L., Area-expanding embeddings of rectangles, arXiv:0710.0403
[4] Guth, L. Isoperimetric inequalities and rational homotopy invariants, arXiv:0802.3550